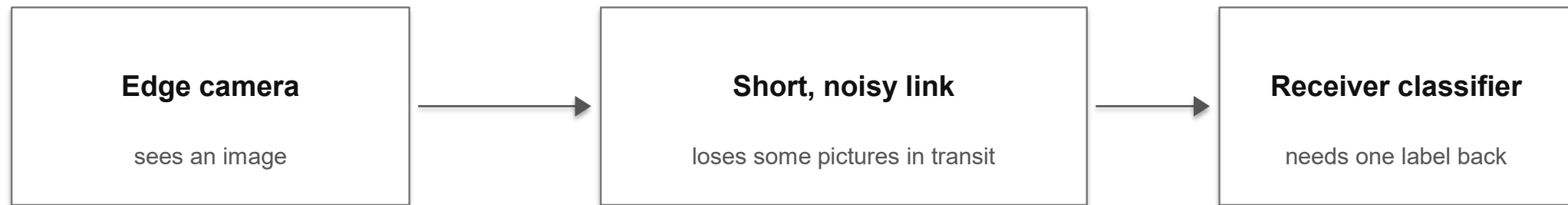


Task-Oriented DeepJSCC Based Semantic Communication

What a camera should send when the receiver only needs the answer



Train the sender and the receiver together so the classification decision survives a short, noisy link.

The thesis

Second Review, 29 September – 3 October 2026.

Progress up to Sep. 30th

All learned models are trained now.

Aug · Digital system	Measured how badly it loses pictures at every setting. All 288,000 held-back combinations tested, and the settings it should use at each signal strength frozen.
Aug · Learned system	Training loop built, models trained at several seeds. Three of them are frozen at 1/6 rate.
Sep · Fair comparison	Built a control that sends learned features digitally, so the comparison includes a task-aware system alongside a compressed image.
Sep · Weak-link training	A second version, with the training signal strength picked at random each time, for comparison against the first.
Sep · Crossing question	Decided it on held-back pictures before seeing the answer: the two lines cross between -5 and -4 dB.
Sep · Power limits	Trained one model under a hard limit on how tall the signal spikes may get.
Sep · Measurements	252 measurements across twelve versions of the system and twenty-one signal strengths, 1,000 held-back pictures each.
Oct · Exam set	Opens in week 11, once the locking document is signed.

Objectives

What we set out to do

- | | |
|-----------------------------------|--|
| 1. Build it | Train a learned sender and receiver together, so the picture goes through the same kind of noise a radio adds. |
| 2. Same radio time | At every signal strength and rate, both systems get the same number of chances to use the radio. |
| 3. Charge the overhead | Checksums, padding and error-correction bytes come out of the budget before any picture data. |
| 4. Pick the setting fairly | The operating point is chosen using the digital system alone, before anyone sees the learned curve. |
| 5. Report per picture | Every row records what happened to one picture, including those that failed to arrive. |

What we measured for this review

We built twelve versions of the system and tested each one at twenty-one signal strengths.

Item	Figure	What it is
Measurements	252	One row each in the published record
Versions of the system	12	Five are controls, not headline systems
Signal strengths	21	From weak to clean
Pictures per point	1,000	The same held-back pictures every time
Exam pictures opened	0	Sealed until week 11

Two things to keep in mind

Radio time	1/6 rate uses the radio 12,800 times per picture; 1/24 rate uses it 3,200 times. Within one rate both systems get exactly the same amount of radio time.
Who marks it	The learned systems are marked by the classifier built into them. The digital systems are marked by one we trained on compressed-looking images. So every comparison in this deck is between whole systems, link and marker included.

The two models behind every number

One classifier reads the pictures. One learned link decides what to transmit.

The learned link

Part	What it is
Encoder	Conv stack, 64 then 128 channels
Blocks	2 residual blocks, Group norm, PReLU
Decoder	Mirror of the encoder, two heads
Heads	Class scores, plus a rebuilt picture
Width sent	8 complex symbols per position, 1,600 positions
Parameters	1.57 M at the 1/6 rate reported here

How it was trained

Setting	Value
Objective	Class scores + 3.0 × reconstruction
Optimiser	Adam, learning rate 0.001
Schedule	Cosine decay, no warm-up
Batch, epochs	32 pictures, 100 epochs
Augmentation	Random crop, horizontal flip
Precision	Mixed 16-bit

The classifier

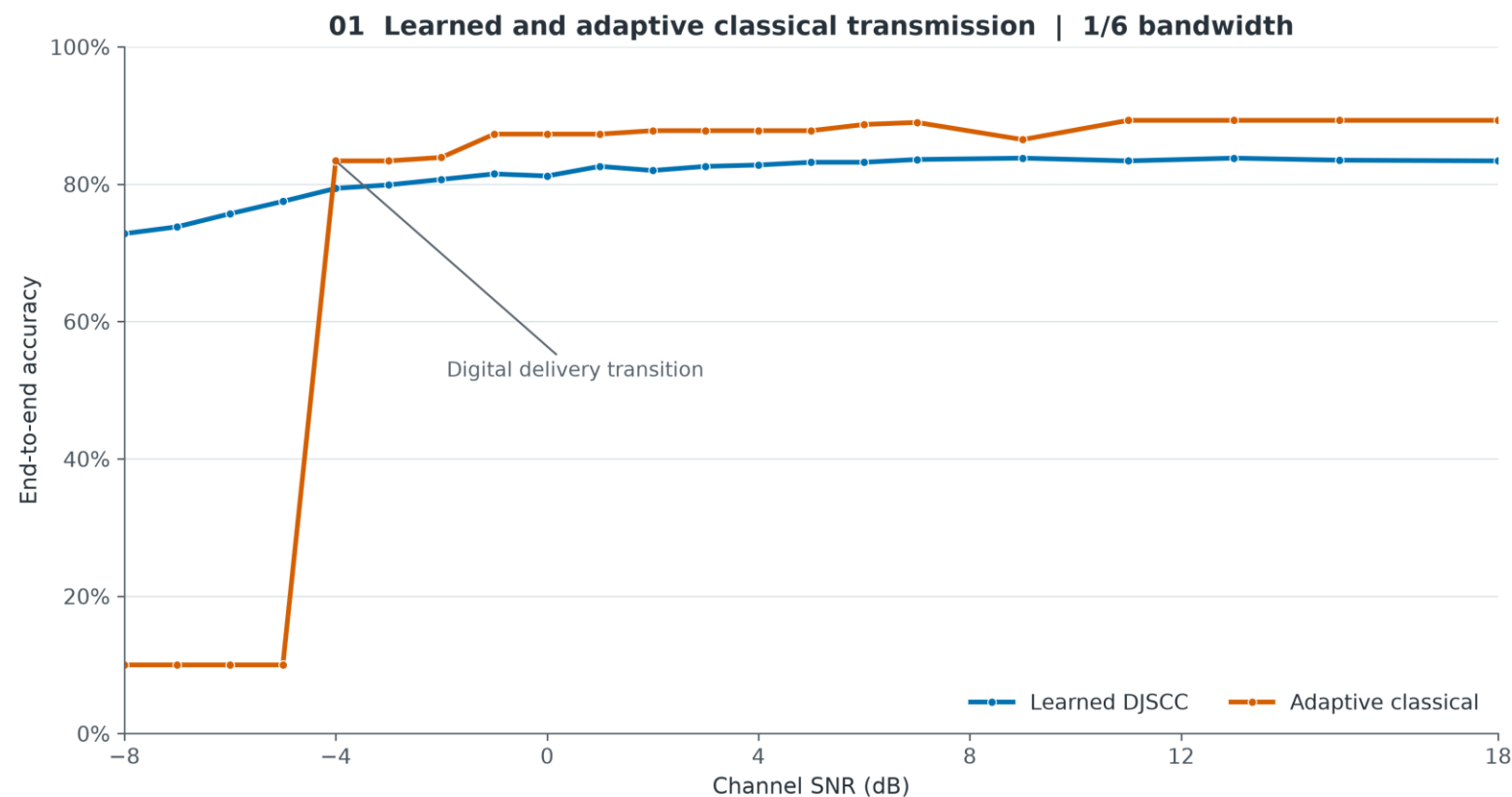
ResNet-18, 11.18 M parameters, trained from scratch. It reads 89.8% of held-back pictures correctly. The digital side is marked by a copy of it retrained on compressed pictures.

Training cost

48.7 seconds per epoch and 1.0 GiB of graphics memory, measured on a laptop GPU.

Where the digital system gives up, and where it takes over

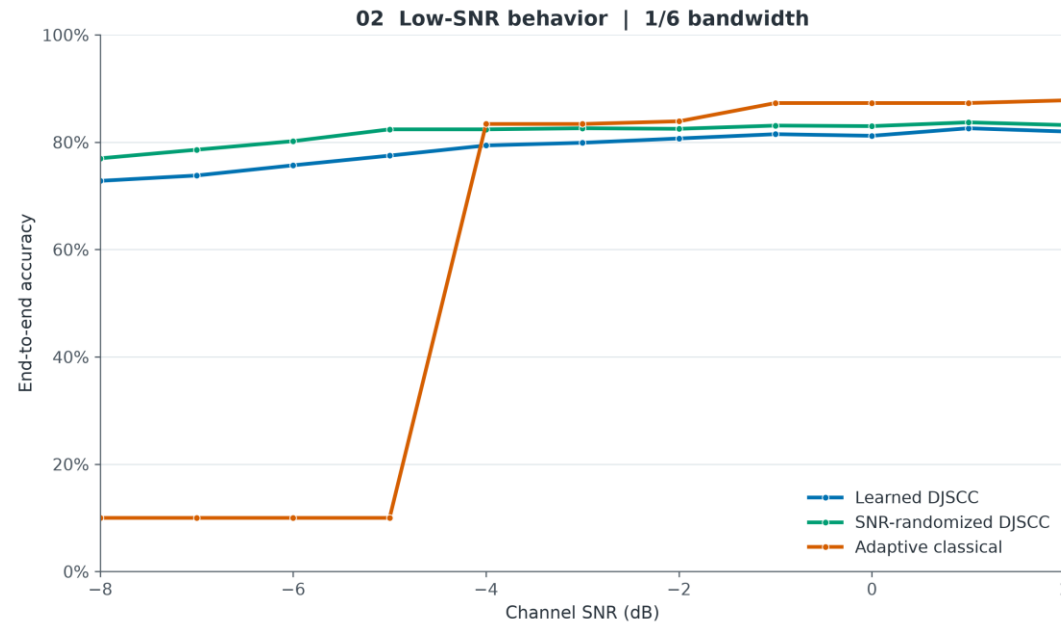
One line runs the whole width of the chart. The other has a step in it near the left.



At the weakest signal the learned system is right on 728 of 1,000 while the digital system delivers nothing. From -4 dB it delivers and goes ahead. At +18 dB it is well ahead.

Close-up: the moment the digital system starts working

Between -5 and -4 dB the digital system begins getting pictures through, and its score jumps 73 points.



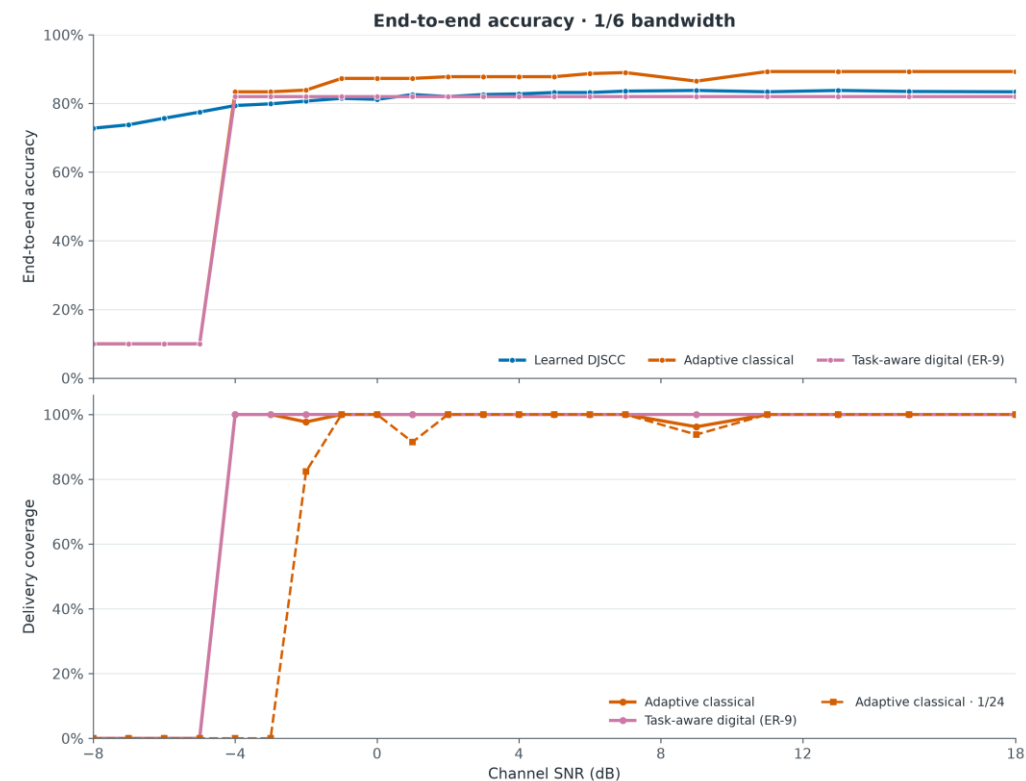
At -8 dB The learned system is right on 728 of 1,000. The digital system got nothing through, so it scores 10%.

At -4 dB The digital system now gets everything through and reaches 83.4%, ahead of the learned system's 79.4%.

After that It dips at -2 dB and again at $+9$ dB.

Why the digital line breaks where it does

The score only moves when the pictures start or stop arriving.



The 10% is a rule

Every picture takes the same fallback answer when delivery fails. On an evenly split set that is right 100 times in 1,000.

Outages are part of it

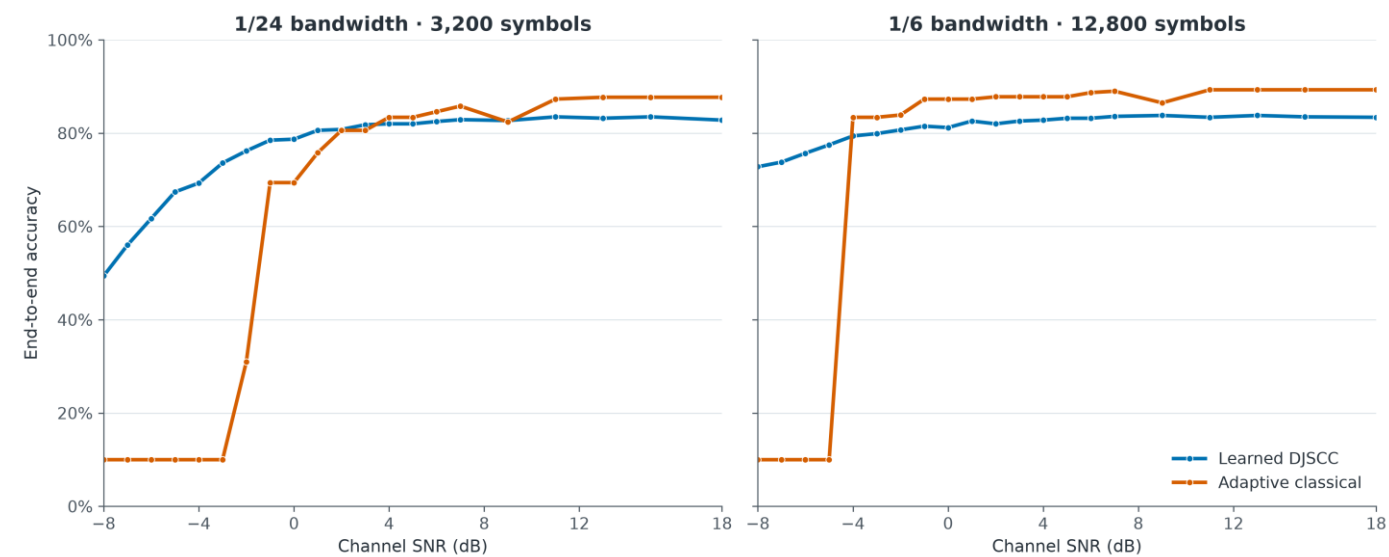
We report how many pictures arrived next to how many we got right, so a broken link is distinguishable from a wrong guess.

Learned always answers

Its rules promise a label at every strength, so its curve never flattens against a floor.

What happens when we cut the radio time fourfold

With four times less radio time, the digital system loses far more at the weak end than the learned one does.



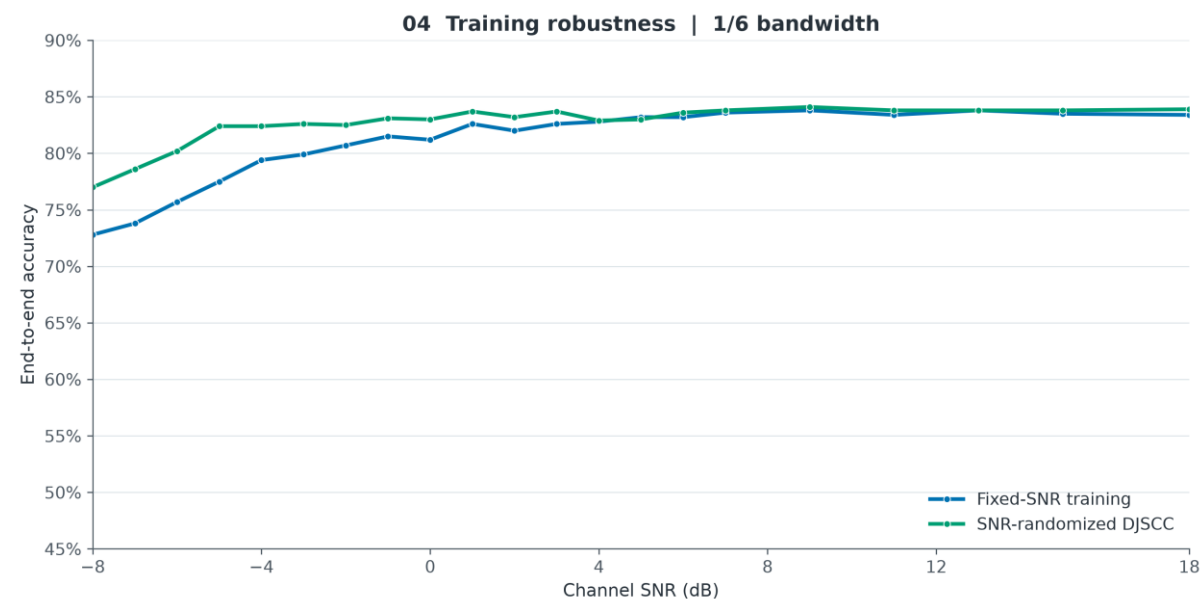
Weak signal Cutting from 1/6 to 1/24 rate drops the learned system from 72.8% to 49.4%.

At 0 dB, 1/24 rate Learned 78.7% against digital 69.4%. The learned system is still ahead.

Catches up, then slips Digital passes learned at the +4 dB point, loses the lead at +9 dB, then goes ahead again.

Training across signal strengths instead of one

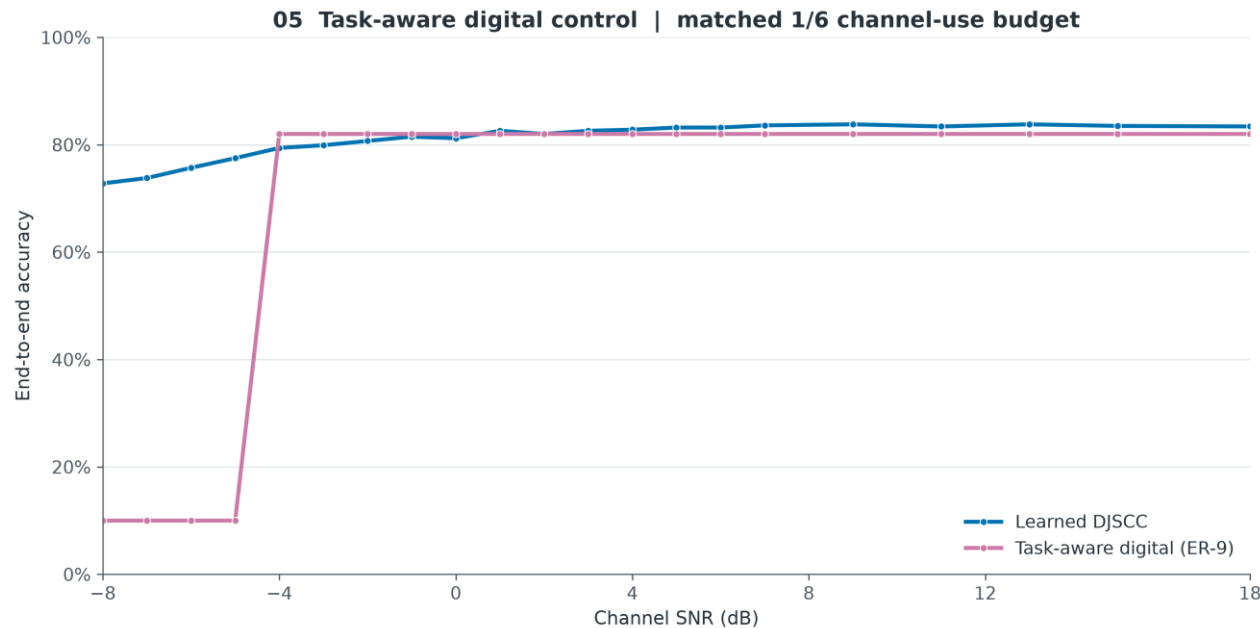
Picking the training signal strength at random each time lifts the whole weak end of the line.



Training	-8 dB	-4 dB	0 dB	+9 dB	+18 dB
Random each time	77.0	82.4	83.0	84.1	83.9
One fixed strength	72.8	79.4	81.2	83.8	83.4

The control that keeps the comparison honest

A compressed image is one kind of digital system. This one sends what the task needs, so it is the harder test.



What this system does

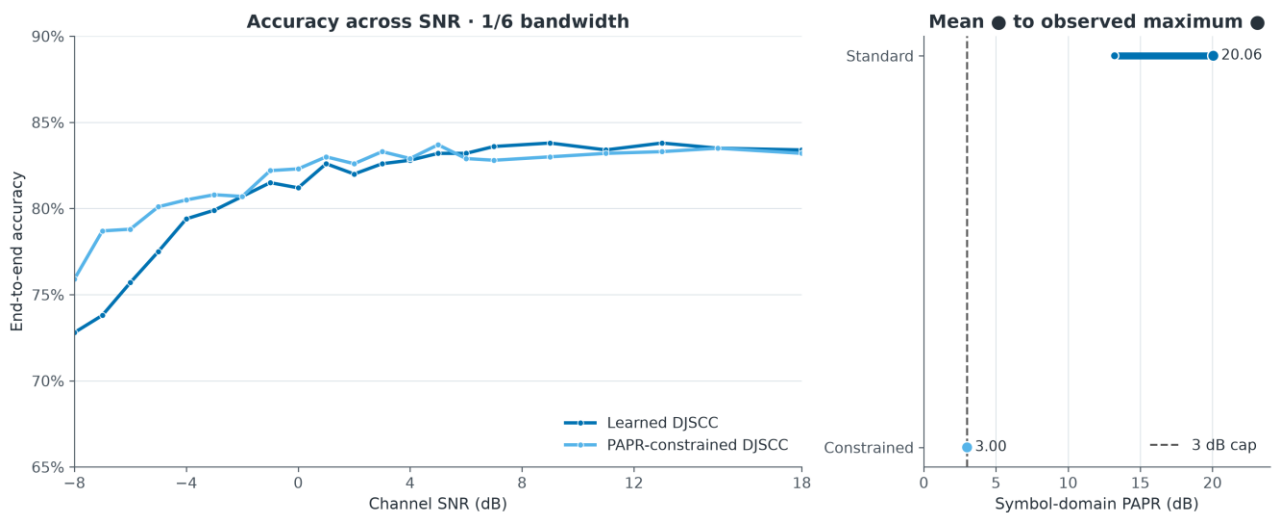
Instead of a JPEG image it sends the learned features themselves, through the same error-correction code and radio settings, using the same 12,800 uses of the radio. It separates “knows the task” from “codes both at once”.

What it settles

It holds 82.0% at every strength from -4 dB up. Sending learned features digitally is strong once pictures get through, so the weak-signal gap is about failed deliveries as much as representation.

Keeping the signal peaks from getting too tall

A real transmitter needs a limit on signal peaks, so we trained a second model under one.

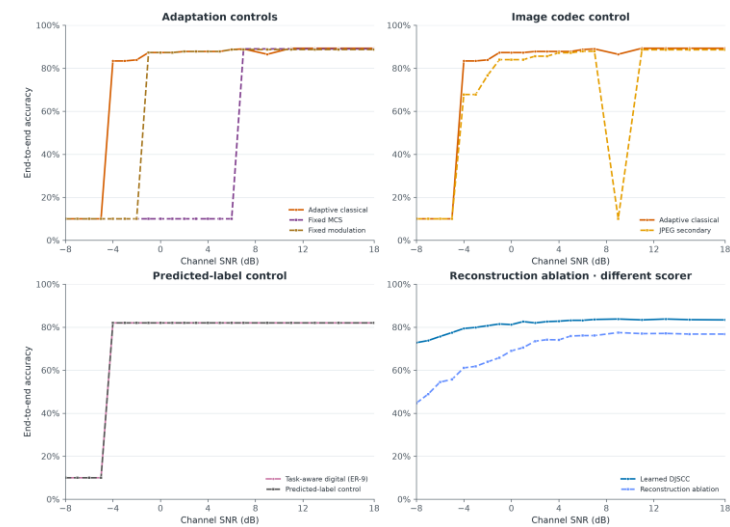


“Peak-to-average power ratio” = how much taller the tallest spike is than the average.

Model	Tallest spike	At -8 dB	At +18 dB
Power-limited	3.0 dB	75.9%	83.2%
Ordinary learned	20.1 dB	72.8%	83.4%

Four controls on the digital system

Each panel takes one design choice away and shows what it was worth.



Fixed settings	Sits at the fallback floor until +6 dB, then reaches 89.0%. Choosing settings per strength buys real reach.
Fixed modulation	Starts working later than adaptive selection, and settles near 88.7%.
JPEG secondary curve	88.6% at +18 dB, with a full outage at +9 dB.
Sending the answer	82.0% once it arrives, using the sender's own predicted label.

The numbers behind the curves

How many of the 1,000 held-back pictures we labelled correctly, one picture at a time.

1/6 rate — 12,800 uses of the radio

System	-8	-5	-4	0	+9	+18
Learned, one strength	72.8	77.5	79.4	81.2	83.8	83.4
Digital, picks settings	10.0	10.0	83.4	87.3	86.5	89.3
Learned, random	77.0	82.4	82.4	83.0	84.1	83.9
Digital, learned feats	10.0	10.0	82.0	82.0	82.0	82.0
Learned, power-limited	75.9	80.1	80.5	82.3	83.0	83.2

1/24 rate — 3,200 uses of the radio

System	-8	-4	-2	0	+4	+18
Learned, one strength	49.4	69.3	76.2	78.7	82.0	82.8
Digital, picks settings	10.0	10.0	31.0	69.4	83.4	87.7

Where the two lines cross

At 1/6 rate, which system is ahead flips between -5 dB and -4 dB. We decided the crossing question on exactly that, before seeing the answer.

Reading the two tables

Each column is one signal strength; each cell is the number of correct labels out of the same 1,000 held-back pictures. At 1/6 rate the learned row leads up to -5 dB and the digital row leads from -4 dB. At 1/24 rate the digital row stays on the fallback floor two dB longer, then climbs past the learned row from +4 dB. The difference between the two tables is the effect of cutting radio time fourfold.

How to read these results

One training run

A single observed measurement per figure, from one run over 1,000 held-back pictures.

Whole systems

Marked by their own classifier on one side, by one trained on compressed images on the other. These numbers describe whole systems.

Different amounts of work

Digital settings and grader were frozen from earlier validation work; the learned versions are separate training runs.

Simulated channel

Spike heights are measured on the numbers we send. Real-radio replay is a stretch goal for late October.

Frozen operating points

The fallback rule and fixed settings produce visible jumps. Every curve passes through measured points.

Both outcomes count as a successful project: a crossing, and the learned system winning everywhere.

Four times we changed the approach because of what we found

Each one was a decision we would have got wrong on paper.

Baseline too weak

We had pinned the digital system to the weakest radio setting, which made the lines crossing impossible by arithmetic. We let it pick a stronger setting as the signal improves.

The rule we adopted: every change strengthens the digital system or is decided in advance.

Signal grid too coarse

Our signal strengths were spaced 2 dB apart exactly where the sharpest drop happens, so we could have missed it entirely. We added three more strengths before it mattered.

Put the detail where the effect is.

A faulty check

The script that checks our packet sizes reported no problems while breaking four of its own rules. We fixed the script, and every result came out the same.

All 215 configurations stayed feasible, and every scientific outcome was unchanged.

Exam set unlocked too early

The exam set was set to unlock in week 9, about three weeks before the document that locks it existed. We moved that to week 11.

They have stayed locked ever since.

Every other change made during the project is recorded, with its reason, in the specification's amendment log.

What we claim as new

The task-aware digital control

A second system sends learned features through the same error-correction code and radio settings, using the same amount of radio time. It tells “knows the task” apart from “codes both at once” inside one experiment.

To our knowledge, classification studies report this gap without a matched control.

Charging the overhead to the digital system

The picture file, framing, checksums and rate-matching bytes are all measured and counted, so the digital system is tuned on the payload it can genuinely use.

Many comparisons quote a nominal rate; here every framing and checksum byte is counted.

What this builds on

- Getting steadily worse as the link worsens
- Avoiding the sudden drop-off
- Sending task-aware features digitally
- Training a neural encoder and decoder

All four are established in the prior literature. This project builds on them.

Negative check

Each claim resolves to a specific entry in the 30-source literature review. The full written novelty statement ships with the final review package.

Where the remaining weeks go

29 Sep – 3 Oct	Second Review	Held-back measurements complete; package and figures frozen	done
5 – 11 Oct	Open the exam set	One guarded opening once the freeze is signed	planned
12 – 18 Oct	Decide everything	Every hypothesis settled on the exam set; first tier frozen	planned
19 – 25 Oct	Demo	Already built; rehearse it with the frozen results	planned
26 Oct – 1 Nov	Poster and hardware	Poster draft; real radio only if the first tier leaves room	planned
2 – 8 Nov	Report freeze	Report finished in the required format	planned
9 – 15 Nov	Spare week	Report completion and checking	planned
16 – 22 Nov	Final Review	Final review 17–21 Nov; report and material due 20 Nov	planned

The report is due on 20 November, inside Final Review week, so the week before it is a freeze.

Where this leaves the project

The learned system keeps answering when the link is too weak for the digital one to get a single picture through. Once the digital system starts delivering, it is the more accurate of the two. They cross between -5 and -4 dB.

Next One guarded run on the exam set once the locking manifest is signed.

Then Every hypothesis settled on the exam set, and the first tier frozen.

Later Real-radio replay and the live demo run once the first tier is frozen.

What we are asking for

Questions on the method or the numbers. The per-picture records behind any point here are available on request. The exam set stays sealed until week 11.